

SAATHI — Team 4

Complete Advanced Intelligence, Cortex, C5–C8 and AI Evaluation Work Package

Operating rules

- Keep the existing ingestion package, versioning package, current AI/backend, and the ~1.25 GB RAGA model. Do not delete or retrain existing assets before benchmarking them.
- Use the existing 600+ MB government-source dataset as the controlled starting corpus. Every source must retain provenance; do not silently add unverified web data.
- Work in separate Git branches and reproducible Colab/Kaggle notebooks. Every experiment must record dataset/model/index version and metrics.
- Never replace a working baseline without a measured comparison.

Mission

- Own the advanced intelligence modules and evaluation layer. This is an independent AI workload, not a team whose only job is integration.

Step 1 — Build the golden evaluation dataset

- Create 75–150 curated questions across standard lookup, product-to-standard, QCO applicability, certification, clause lookup, revision comparison, scheme selection, compliance, Hindi, unknown and false-premise cases.
- For each record store expected evidence, facts, clause/page where available, language, difficulty and failure type.
- Version the dataset so metrics remain comparable.

Step 2 — Knowledge graph / compliance chain

- Create structured entities for product, standard, QCO, scheme, test, lab and document.
- Represent relationships with evidence references and verification status.
- If a relationship cannot be established, return 'Not verified from available sources' rather than an inferred relationship.

Step 3 — C5 Scheme intelligence

- Build product → standard → QCO → scheme mapping using deterministic rules plus source evidence.
- Use the RAGA model only to explain a verified route.
- Test unsupported product categories and conflicting standards.

Step 4 — C6 Compliance chain

- Build the source-backed chain from product through standard, QCO, scheme, tests, lab and required documents.
- Every edge must have evidence or an explicit unverified state.
- Create chain validation tests.

Step 5 — C7 Laboratory matching

- Create deterministic filters for standard, required test, product and other verified lab parameters.
- Rank matching labs only from authoritative data.
- Use AI to explain the match; never invent accreditation, capabilities or official status.

Step 6 — C8 Regulatory change intelligence

- Detect changes from the controlled source dataset.
- Link a changed source to affected standards/QCOs and requirements.
- Generate impact summaries only from detected evidence.
- Test old/new versions and amendment scenarios.

Step 7 — Cortex document compliance

- Test uploaded document → extraction → evidence retrieval → requirement matching → deterministic rules → LLM explanation.
- Distinguish PASS, FAIL and MISSING EVIDENCE.
- Every finding must cite the document evidence and the applicable requirement.

Step 8 — RAG evaluation dashboard

- Calculate Recall@K, Precision@K, MRR, nDCG, citation correctness, groundedness, unsupported-answer rate, retrieval latency, LLM latency and failure rate.
- Clearly separate benchmark results from demo telemetry.
- Create a regression report after every model/retrieval change.

Step 9 — Adversarial testing

- Create deliberately misleading questions, false premises, ambiguous products, conflicting standards, old-version queries and unsupported claims.
- Track whether the system declines, asks for clarification or identifies the correct evidence.

Deliverables

- golden evaluation dataset
- evaluation runner
- metrics report
- compliance knowledge graph prototype
- C5/C6/C7/C8 test modules
- Cortex validation suite
- adversarial test suite
- RAG quality dashboard data

Acceptance test

- All major RAG changes have measurable before/after results.
- C5–C8 relationships are evidence-backed.
- Cortex distinguishes missing evidence from failure.
- Regression tests prevent improvements in one area from silently breaking another.